LAB RECORD

Computational Thinking with Python

DSC – C

Submitted By

Anagha Bilwadala S

Dept. of Physics

INDEX

Sl.No.	DATE	PROGRAM	PAGE NO
1	30/08/2024	DISPLAY NAME	3
2	30/08/2024	SUM OF TWO NUMBERS	5
3	30/08/2024	LARGEST OF TWO NUMBERS	7
4	04/10/2024	CHECKING IF THE NUMBER IS EVEN OR ODD	9
5	04/10/2024	GUESSING THE NUMBER	11
6	18/10/2024	ONLINE STORE	13
7	18/10/2024	PIZZAORDERING PROCESS	15
8	18/10/2024	MY PETS	17
9	25/10/2024	ARRAY	19
10	25/10/2024	FACTORIAL OF A NUMBER	21
11	08/11/2024	CALCULATOR	23

SOURCE CODE

```
name=input("Enter your name: ")
print(name)
```

Output

Enter your name: Anagha

Anagha

PROGRAM -1 **30/08/2024**

AIM:

Program to display name

ALGORITHM:

Step1:Start

Step2:Print a message asking for username

Step3:Store the user's input in a variable

Step4:Print the users name

Step5:Stop

Result

Code executed successfully

SOURCE CODE

```
a=int(input("Enter the first number: "))
b=int(input("Enter the second number: "))
sum=a+b
print(sum)
```

Output

Enter the first number: 3

Enter the second number: 5

8

PROGRAM 2 **30/08/2024**

AIM:

Program to display sum of two numbers

ALGORITHM:

Step 1:Start

Step 2:Input 1st number as a

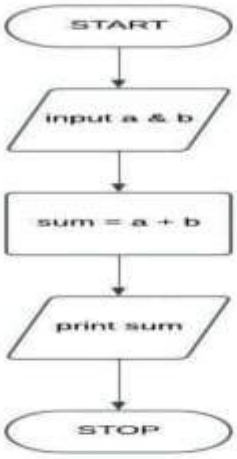
Step 3:Input 2nd number as b

Step 4:Define sum=a+b

Step 5:Print sum

Step 6:Stop

FLOWCHART:



RESULT: Code Executed Successfully

SOURCE CODE

```
num1=int(input("Enter the first number: "))
num2=int(input("Enter the second number: "))
if num1>num2:
  print("num1 is greater than num2")
else:
  print("num2 is greater than num1")
```

Output

Enter the first number: 6

Enter the second number: 4

num1 is greater than num2

Enter the first number: 4

Enter the second number: 5

num2 is greater than num1

PROGRAM 3 **30/08/2024**

AIM:

Program to find the largest of two numbers

Algorithm

Step 1:Start

Step 2:Input the 2 numbers as a and b

Step 3: If num1>num2

Go to step4

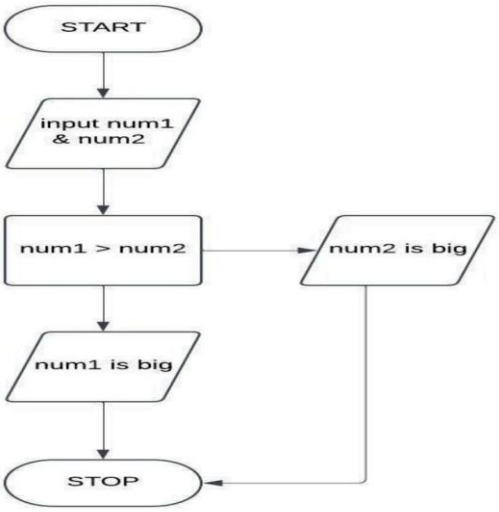
Else go to step5

Step 4: Print num1 is greater

Step 5:Print num2 is greater

Step 6: Stop

FLOWCHART:



RESULT:

Code executed succesfully

SOURCE CODE

```
n=int(input("Enter your number: "))
if 1<=n<=100:
  if n%2==0:
    print("n is between 1 and 100 and is even")
  else:
    print("n is between 1 and 100 and is odd")
else:
  print("n is not between 1 and 100")
```

Output

Enter your number: 3

n is between 1 and 100 and is odd

Enter your number: 200

n is not between 1 and 100

Enter your number: 2

n is between 1 and 100 and is even

PROGRAM 4 **04/10/2024**

AIM:

Program to check if a number is between 1 and 100 and is even or odd

Algorithm

Step 1: Start

Step 2: Input a number as n

Step 3: If n≤100

Go to step 5

Else go to Step 4

Step 4: Print the number is not between 1 and 100

Step 5:if n/2=0,

Go to Step 6

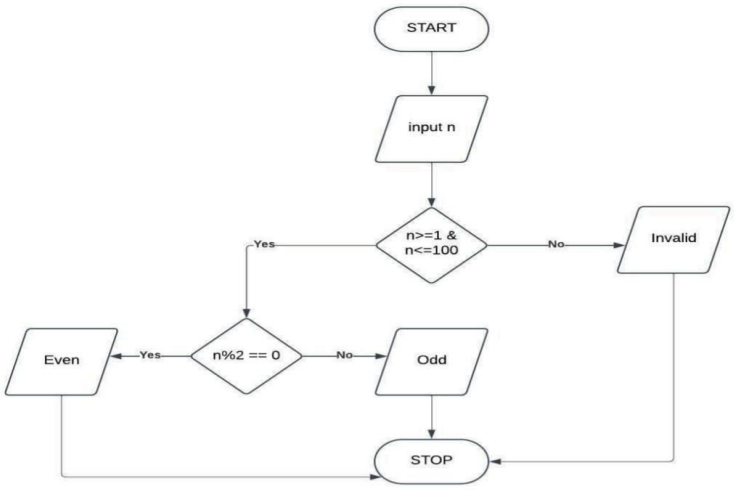
Else go to step 7

Step 6: Print the number is even

Step 7: Print the number is odd

Step 8:Stop

Flowchart:



Result:

Code executed successfully

SOURCE CODE

```
import random
a = random.randint(1, 10)
i = 5
for number in range(5):
    b = int(input("Enter a number between 1 and 10: "))

    if a == b:
        print("You won!")
        break

    elif b > a:
        i -= 1
        print(f"The number is too large. You have {i} chances left.")

    elif b < a:
        i -= 1
        print(f"The number is too small. You have {i} chances left.")
    if i == 0:
        print("You lose! The correct number was "+str(a)+".")
        break
```

Output

Enter a number between 1 and 10: 4

The number is too small. You have 4 chances left.

Enter a number between 1 and 10: 8

The number is too small. You have 3 chances left.

Enter a number between 1 and 10: 9

You won!

PROGRAM 5 **04/10/2024**

AIM:

Program to select a number between 1 and 10 by giving 5 guesses

Algorithm

Step 1: Start

Step 2: Generate a random number

Step 3:Set the number of guesses 5

Step 4:Loop to give the player 5 guesses.

Step 5:If user input= computer number

Step 6: Display you win

Step 7: Else display you lose

Step 8: Stop

Result:

Code run successfully

SOURCE CODE

```
print("Welcome to online_store")
online_store={
    't_shirt':150,
    'bottle':25,
    'keychain':10
}
print(online_store)
x=input("Enter the item you want to buy: ")
if x not in online_store:
   print("Item is not available")
else:y=int(input("Enter the number of items you want to buy: "))
total_price=online_store[x]*y
print("You chose to buy "+str(y)+" "+str(x))
if y>=10:
  total_amount=total_price-(total_price*0.1)
  print("The total amount after discount is"+str(total_amount))
else:
  print("The total amount is "+str(total_price))
print("Thank you for shopping with us")
```

Output

Welcome to online_store

{'t_shirt': 150, 'bottle': 25, 'keychain': 10}

Enter the item you want to buy: t_shirt

Enter the number of items you want to buy: 10

You chose to buy 10 t_shirt

The total amount after discount is 1350.0

Thank you for shopping with us

PROGRAM 6 **18/10/2024**

AIM:

Program to simulate an online store purchase process

Algorithm

Step 1:Start

Step 2: Create a dictionary with items and price

Step 3: Input the item customer wants to buy and in a variable x

Step 4: If x not in online store

Print that the item is not available

Else go to Step 5

Step 5: Input the number of items in a variable y

Step 6:Calculate total price = online_store[x]*y

Step 7: If y≥10

Apply discount

total amount=total price-(total price*0.1)

and print the total amount

else go to Step 8.

Step 8: Display total price

Step 9: Stop

Result:

Code executed successfully

SOURCE CODE

```
print("Welcome to Pizza Palace!")
size = input("Choose your pizza size (personal/family): ")
if size == 'personal':
    print("You have selected Personal Pizza")
elif size == 'family':
    print("You have selected Family Pizza")
else:
    print("Invalid input. Please enter 'personal' or 'family'.")
    exit()
toppings = input("What topping do you want?\nEnter 'm' for mushrooms, 'o' for onion, 'c' for chicken: ")
if toppings == "m":
    print("You have selected mushrooms as toppings")
elif toppings == "o":
    print("You have selected onion as toppings")
elif toppings == "c":
    print("You have selected chicken as toppings")
else:
    print("Invalid input. Please enter 'm', 'o', or 'c'.")
    exit()
cheese = input("Do you want extra cheese? Enter 'y' for yes and 'n' for no: ")
if cheese == "y":
    print("You have selected extra cheese")
elif cheese == "n":
    print("You have not selected extra cheese")
else:
    print("Invalid input. Please enter 'y' or 'n'.")
    exit()
print("Your order is confirmed. Thank you for dining with us!")
```

Output

Welcome to Pizza Palace!

Choose your pizza size (personal/family): personal

You have selected Personal Pizza

What topping do you want?

Enter 'm' for mushrooms, 'o' for onion, 'c' for chicken: m

You have selected mushrooms as toppings

Do you want extra cheese? Enter 'y' for yes and 'n' for no: y

You have selected extra cheese

Your order is confirmed. Thank you for dining with us!

PROGRAM 7 **18/10/2024**

AIM:

Program for pizza ordering process

Algorithm

Step 1: Start

Step 2: Print a Welcome message

Step 3: Ask the user to choose the pizza size (personal or family)

Step 4: If the input is “personal”, print conformation

If the input is “family”, print conformation

If the input is invalid, print error message and end

Step 5: Ask the user to choose a topping

(‘m’ for mushrooms, ‘c’ for chicken and ‘o’ for onion)

Step 6: If the input is “m”, print conformation

If the input is “o”, print conformation

If the input is “c”, print conformation

If the input is invalid, print error message and end.

Step 7: Ask the user if they want extra cheese

(‘y’ for yes and ‘n’ for no)

Step 8: If the input is “y”, print confirmation

If the input is “n”, print confirmation

If the input is invalid, print an error message and end.

Step 9: Print the order confirmation message

Step 10: Stop

Result

Code executed successfully

SOURCE CODE

```
def myPets(cat,dog,bird):
  print(f"The Cat's name is "+cat+" The Dog's name is "+dog+" and the bird's name is "+bird+".")
myPets("Whiskers","KingKong","Pirate")
myPets("Kitty","Max","Tweety")
myPets("Sophie","Tiger","Pinky")
```

Output

The Cat's name is Whiskers The Dog's name is KingKong and the bird's name is Pirate.

The Cat's name is Kitty The Dog's name is Max and the bird's name is Tweety.

The Cat's name is Sophie The Dog's name is Tiger and the bird's name is Pinky.

PROGRAM 8 **18/10/2024**

AIM:

Program for a function that accepts arguments

Algorithm

Step 1: Start

Step 2: Define the function myPets

Step 3: Print a sentence starting with the names of each pet using provided data

Step 4: Stop

Result

Code executed successfully

SOURCE CODE

```
import numpy as np
arr=np.array([1,2,3,4,5])
print(arr)
```

Output

[1 2 3 4 5]

PROGRAM 9 **25/10/2024**

AIM:

Program to display array using numpy

Algorithm

Step 1: Start

Step 2: Import numpy

Step 3: Create an array

Step 4:Print the array

Step 6: Stop

Result

Code executed successfully

Input

```
def fact(n):
  if n==0:
    return 1
  else:
    return n*fact(n-1)
n=int(input("Enter a number: "))
print(fact(n))
```

Output

Enter a number: 3

6

PROGRAM 10 **25/10/2024**

AIM:

Program to find the factorial of a number

Algorithm

Step 1: Start

Step 2:Define factorial

Step 3: Set the base case for recursion: if n=0 or n=1, return 1

Step 4: Recursively call the function

For n>1, return n*factorial(n-1)

Step 5:Input number from user

Step 6: Print result

Step 7: Stop

Result

Code executed successfully

SOURCE CODE

```
def add(x, y):
return x+y
def substract (x, y):
return x-y
def multiply(x, y):
return x*y
def divide(x, y):
return x/y
pr int ("Which Option do you want to choose")
pr int ("1. Addition")
pr int ("2. Substraction")
pr int ("3. Multiplication")
pr int ("4. Divi s ion")
choice = input ("enter your choice(1/2/3/4)")
num1 = int (input ("enter fi r s t number "))
num2 = int (input ("enter second number "))
if choice == "1":
print ("Result : ", add(num1, num2))
elif choice == "2":
pr int ("Result: ", subs t ract (num1, num2))
elif choice == "3":
pr int ("Result: ", mul t iply(num1, num2))
elif choice == "4":
print ("Result: ", divide(num1, num2))
else:
print ("Invalid input ")
```

OUTPUT:

Which Option do you want to choose

1. Addition

2. Substraction

3. Multiplication

4. Division

enter your choice(1/2/3/4)3

enter first number5

enter second number6

Result: 30

PROGRAM-11 **08/11/2024**

AIM:

Program to create a calculator

Algorithm

STEP 1: START

STEP 2: Define Functions for Operations

STEP 3: Display Menu of Operations

STEP 4: Get User Choice

STEP 5: Input Number

STEP 6: Perform Operation Based on User Choice

STEP 7: Display the results

STEP 8: STOP

RESULT:

Code Executed Successfully